# An exact three-copy undistillability bound

Uniform in local dimension, at $\alpha = 462713/1000000$

**Abstract**

We give an exact-arithmetic, computer-assisted proof that Werner states with parameter $0 \le \alpha \le 0.462713$ are three-copy undistillable in every finite local dimension. The proof combines a dimension-independent quadratic inequality for arbitrary complex rank-two matrices with a rational four-replica certificate. Positivity of 238 strictly positive leading blocks implies positivity in all sufficiently large dimensions; isometric embedding then covers every finite dimension. The analytic inequality is proved here. The accompanying rational data and independent checker comprise the finite computational part of the proof.

## 1 Statement and conventions

Let $F(x \otimes y) = y \otimes x$ on $\mathbb{C}^d \otimes \mathbb{C}^d$, and define

$$\rho_{d,\alpha} = \frac{I - \alpha F}{d(d-\alpha)}, \qquad \alpha_0 = \frac{462713}{1000000} = 0.462713. \tag{1}$$

All partial traces below are unnormalized. For a matrix $X$ on $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3$, write

$$Q_\alpha(X) = \sum_{S \subseteq \{1,2,3\}} (-\alpha)^{|S|} \| \operatorname{Tr}_S X\|_F^2 = N - \alpha B_1 + \alpha^2 B_2 - \alpha^3 T, \tag{2}$$

where $N = \|X\|_F^2$, $B_j = \sum_{|S|=j} \| \operatorname{Tr}_S X\|_F^2$, and $T = |\operatorname{Tr} X|^2$. A missing second singular value is taken to be zero.

**Theorem 1.** *For arbitrary finite-dimensional complex spaces $\mathcal{H}_1, \mathcal{H}_2, \mathcal{H}_3$ and every matrix $X$ on their tensor product with* $\operatorname{rank} X \le 2$*,*

$$Q_{\alpha_0}(X) \ge 0.$$

*Consequently, for every integer $d \ge 2$ and $0 \le \alpha \le \alpha_0$, $\rho_{d,\alpha}$ is three-copy undistillable. It has negative partial transpose when $\alpha > 1/d$; in particular the theorem gives a nonempty NPT interval $1/d < \alpha \le \alpha_0$ for every $d \ge 3$.*

Here three-copy undistillability is the Schmidt-rank-two condition

$$\langle \psi | (\rho_{d,\alpha}^{\Gamma})^{\otimes 3} | \psi \rangle \ge 0 \quad \text{for every } \psi \text{ of Schmidt rank at most two,}$$

with the three copies grouped by party. Indeed $F^\Gamma = |\Omega_d\rangle\langle\Omega_d|$ for $\Omega_d = \sum_{a=1}^d e_a \otimes e_a$, and vectorization identifies Schmidt rank with matrix rank and the expectation with $Q_\alpha(X)/[d(d-\alpha)]^3$. A Schmidt-rank-two vector is the pullback of a two-qubit vector under local filters, giving the usual two-qubit-output criterion for distillation. The eigenvalue $1 - d\alpha$ of $I - \alpha|\Omega_d\rangle\langle\Omega_d|$ gives the asserted NPT condition. The theorem concerns three physical copies, not arbitrary copy number.

## 2 A quadratic inequality and its determinant

The analytic ingredient is the following inequality, proved in full in Appendix A:

$$\mathcal{A}(X) := \frac{17}{8} N - B_1 + \frac{1}{2} B_2 - \frac{5}{16} T \;\ge\; \frac{(\sigma_1 - \sigma_2)^2}{4} \ge 0 \qquad (\operatorname{rank} X \le 2). \tag{3}$$

It applies to arbitrary complex $X$ and unequal factor dimensions.

For four vectors $r = (u_0, v_0, u_1, v_1)$ in $\mathcal{H}$, put $D_j = u_j v_j^\dagger$. For real subset weights $w = (w_S)$ define

$$(H_w)_{ij} = \sum_S w_S \langle \operatorname{Tr}_S D_i, \operatorname{Tr}_S D_j \rangle_F.$$

Then $c^\dagger H_w c = \sum_S w_S \| \operatorname{Tr}_S(c_0 D_0 + c_1 D_1)\|_F^2$. Every combination $c_0 D_0 + c_1 D_1$ has rank at most two. Thus (3) implies $H_a \succeq 0$ and $\det H_a \ge 0$ for the weights

$$a_\varnothing = 17/8, \quad a_S = -1 \ (|S| = 1), \quad a_S = 1/2 \ (|S| = 2), \quad a_{123} = -5/16. \tag{4}$$

We make the polynomial determinant explicit. Replicas are ordered $(u_0, v_0, u_1, v_1)$ and numbered $0, 1, 2, 3$. A permutation $p \in S_4$ maps input position $j$ to output position $p(j)$; products act right to left. For $e, f \in \{0, 1\}$ set

$$d_{ef} = (01)^e(23)^f, \qquad c_{ef} = (03)^e(12)^f(02)(13).$$

Writing $e_i = 1_{i\in S}$ and $f_i = 1_{i\in R}$, define the formal tensor

$$K(w) = \sum_{S,R\subseteq\{1,2,3\}} w_S w_R \left( \bigotimes_{i=1}^{3} P_{d_{e_i f_i}} - \bigotimes_{i=1}^{3} P_{c_{e_i f_i}} \right). \tag{5}$$

Regrouping the four replicas by physical site, contraction of indices gives

$$\langle u_0 \otimes v_0 \otimes u_1 \otimes v_1, K(w)(u_0 \otimes v_0 \otimes u_1 \otimes v_1)\rangle = \det H_w. \tag{6}$$

The first permutation product gives $(H_w)_{00}(H_w)_{11}$ and the second gives $(H_w)_{01}(H_w)_{10}$. Let $K_\alpha = K(w)$ with $w_S = (-\alpha)^{|S|}$.

Let $\mathcal{S}$ denote averaging over physical-site permutations, simultaneous conjugation of replica labels by

$$V_4 = \{1, (01)(23), (02)(13), (03)(12)\},$$

and adjoint. Put $C = \mathcal{S}K(a)$. These operations preserve four-vector products and nonnegativity of their real expectations, so

$$\langle r, Cr\rangle \geq 0 \quad \text{for every four-vector product } r. \tag{7}$$

This does not assert positivity of $C$ on arbitrary entangled vectors of the four-replica space.

# 3 The rational coefficient certificate

Use the eight transpose sets

$$\mathcal{T} = \{\varnothing, \{0\}, \{1\}, \{2\}, \{3\}, \{0, 1\}, \{0, 2\}, \{0, 3\}\}.$$

The supplied rational arrays define $Y_\tau = \sum_{\mathbf{p}\in S_4^3} y_{\tau,\mathbf{p}} P_{p_1} \otimes P_{p_2} \otimes P_{p_3}$ and a tensor $E$ satisfying the exact coefficient identity

$$K_{\alpha_0} = \sum_{\tau\in\mathcal{T}} Y_\tau + E + \mu C, \qquad \mu = \frac{1581841004766715}{18014398509481984} > 0. \tag{8}$$

All coefficients are rational constants independent of dimension. The tensor $E$ is supported only on monomials for which one of

$$p_i(0) = 2, \quad p_i(2) = 0, \quad p_i(1) = 3, \quad p_i(3) = 1$$

holds for all three sites $i$, with the same choice throughout. Each such monomial contains one of the full-vector inner products $\langle u_1, u_0\rangle$, $\langle u_0, u_1\rangle$, $\langle v_1, v_0\rangle$, $\langle v_0, v_1\rangle$ as a factor. It therefore vanishes on orthogonal left and right singular frames. There are 819 such monomials, forming 40 symmetry orbits.

Write $\Gamma_\tau$ for transpose on the full replica factors in $\tau$, at all three physical sites. If

$$\Gamma_\tau(Y_\tau) \succeq 0 \quad (\tau \in \mathcal{T}), \tag{9}$$

then $\langle r, Y_\tau r\rangle = \langle r^{*\tau}, \Gamma_\tau(Y_\tau) r^{*\tau}\rangle \geq 0$; the selected full vectors are conjugated in $r^{*\tau}$. Equations (7)–(9) imply $\det H_{w(\alpha_0)} \geq 0$ on every orthogonal singular frame.

The diagonal entries are positive for nonzero dyads: with $F_i$ denoting the local swaps between two replicas,

$$Q_\alpha(uv^\dagger) = \langle u \otimes v, \prod_{i=1}^{3}(I - \alpha F_i)(u \otimes v)\rangle \geq (1 - \alpha)^3 \|u\|^2 \|v\|^2 \qquad (0 \leq \alpha < 1). \tag{10}$$

The swaps commute and each factor is bounded below by $(1 - \alpha)I$. Thus the determinant inequality makes this two-by-two Hermitian restriction positive semidefinite. Applying it to the SVD of an arbitrary rank-two $X$ proves $Q_{\alpha_0}(X) \geq 0$ whenever (9) holds.

# 4 Certification in every dimension

## Complete polynomial representations

At one physical site, the 24 operators $D_{\tau,p} = \Gamma_\tau(P_p)$ have pairing diagrams with four input and four output vertices. For $P_p$, input $j$ is paired with output $p(j)$; transpose exchanges the input and output vertices in $\tau$. A matrix entry is the product of the Kronecker deltas along the four edges. Composition is concatenation of diagrams, with one factor $d$ per closed loop. Adjoint reflects the diagram; trace closes matching input/output vertices.

For $d \geq 4$, these 24 operators are linearly independent: assign four distinct labels to the edges of any selected diagram. This matrix entry is one for that diagram and zero for every other pairing. Their physical algebra therefore has dimension 24.

The second data file supplies matrices $\rho_{\tau,r}(p;d)$ and symmetric Grams $G_{\tau,r}(d)$ over $\mathbb{Q}[d]$, and polynomial multiplicities $m_{\tau,r}(d)$. The checker verifies the following as polynomial identities:

$$\rho_r(p)\rho_r(q) = d^{\ell(p,q)}\rho_r(p\cdot q), \qquad \rho_r(p)^T G_r = G_r\rho_r(p^*), \tag{11}$$

$$\mathrm{Tr}_{\mathrm{phys}}\, D_{\tau,p} = \sum_r m_{\tau,r}(d)\,\mathrm{tr}\,\rho_{\tau,r}(p;d), \qquad \sum_r (\dim \rho_{\tau,r})^2 = 24. \tag{12}$$

Here $p\cdot q$ labels the concatenated diagram and $\ell(p,q)$ counts its closed loops; for a nonempty transpose set this is not generally ordinary permutation multiplication. It also checks $\rho_r(1) = I$, $G_r(d) \succ 0$ and $m_r(d) > 0$ for every real $d \geq 4$. Positivity follows by substituting $d = 4 + u$: every Gram leading principal minor and multiplicity has nonnegative rational coefficients and a positive constant term.

These checks prove completeness. The map $\Phi(a) = \bigoplus_r \rho_r(a)$ is a homomorphism. If $\Phi(a) = 0$, (12) applied to $a^\dagger a$ gives $\mathrm{Tr}_{\mathrm{phys}}(a^\dagger a) = 0$, hence $a = 0$. Domain and codomain both have dimension 24, so $\Phi$ is an isomorphism. Conjugating each block by $G_r^{1/2}$ makes it a $*$-isomorphism. Physical positivity is therefore equivalent to positivity in every supplied block. No irreducible component is left unchecked.

## Strict leading blocks

For each block, let $k_r$ be the largest degree in $d$ of any entry of any $\rho_r(p;d)$, and let $g_r$ be the largest degree of a Gram entry. With $t = 1/d$, set

$$R_{r,p}(t) = t^{k_r}\rho_r(p;1/t), \qquad \overline{G}_r(t) = t^{g_r}G_r(1/t).$$

Both are polynomial; the checker verifies $\overline{G}_r(0) \succ 0$. The complete degree data are:

| Transpose representative | Block orders | $k_r$ | $g_r$ | Triple blocks |
|---|---|---|---|---|
| $\varnothing$ | $1,1,2,3,3$ | $0,0,0,0,0$ | $0,0,0,0,0$ | 35 |
| $\{0\}$ | $1,1,2,3,3$ | $0,0,0,1,1$ | $0,0,0,1,1$ | 35 |
| Each of $\{0,1\},\{0,2\},\{0,3\}$ | $1,1,1,1,2,4$ | $0,0,0,0,2,1$ | $0,0,0,0,1,1$ | 56 |

For a triple $\mathbf{r} = (r_1, r_2, r_3)$ define, using the same raw coefficients as in (8),

$$M_{\tau,\mathbf{r}}(t) = \sum_{p_1,p_2,p_3} y_{\tau,\mathbf{p}} \bigotimes_{j=1}^{3} R_{\tau,r_j,p_j}(t), \tag{13}$$

$$B_{\tau,\mathbf{r}}(t) = \left(\bigotimes_j \overline{G}_{\tau,r_j}(t)\right) M_{\tau,\mathbf{r}}(t). \tag{14}$$

These are polynomial, and $B$ is symmetric by (11) and the verified Hermitian coefficient symmetries. At positive $t = 1/d$, $B$ is the actual Gram-weighted physical block of $\Gamma_\tau(Y_\tau)$ multiplied by the positive scalar $t^{\sum_j (k_{r_j}+g_{r_j})}$. This changes only how a block is tested, not its coefficients or positivity.

For every representative triple, the exact certificate satisfies

$$G_0(M_0 - \varepsilon I) \succeq 0, \qquad \varepsilon = \frac{1}{2000000000}, \qquad G_0 = \bigotimes_j \overline{G}_{r_j}(0), \quad M_0 = M_{\tau,\mathbf{r}}(0). \tag{15}$$

Here $I$ is the matrix identity, not the leading image of the identity diagram, which may vanish. Since $G_0 \succ 0$, (15) implies $B(0) \succeq \varepsilon G_0 \succ 0$.

The coefficient symmetries identify reordered triples of block labels; $V_4$ identifies the four singleton transpose sets. Complementary transpose sets differ by full transpose, preserving positivity. Consequently the complete list comprises $\binom{7}{3} + \binom{7}{3} + 3\binom{8}{3} = 238$ representative blocks, of order at most 64. There are finitely many continuous matrices $B(t)$, all strictly positive at zero. They are therefore simultaneously positive for all sufficiently small positive $t$. Thus (9) holds for every sufficiently large integer $d$.

### Descent and the parameter interval

Given arbitrary factor dimensions $d_i$, choose a sufficiently large common dimension $D \geq \max_i d_i$ and isometries $V_i : \mathcal{H}_i \to \mathbb{C}^D$. For $\widetilde{X} = (\bigotimes_i V_i) X (\bigotimes_i V_i)^\dagger$,

$$\mathrm{Tr}_S \widetilde{X} = (\bigotimes_{i \notin S} V_i)(\mathrm{Tr}_S X)(\bigotimes_{i \notin S} V_i)^\dagger.$$

Rank and all partial-trace Frobenius norms are preserved. Hence $Q_{\alpha_0}(X) = Q_{\alpha_0}(\widetilde{X}) \geq 0$. An explicit dimension cutoff is unnecessary: an arbitrarily large certified dimension can be chosen for each proposed finite-dimensional matrix.

For completeness the lower-parameter interval follows without any dimension-dependent estimate. Put $A_\alpha = I - \alpha |\Omega_d\rangle\langle\Omega_d|$. Schmidt-rank-two positivity of $A_{\alpha_0}^{\otimes 3}$ implies that of $A_{\alpha_0}^{\otimes k}$ for $k \leq 3$, by adjoining product vectors with positive spectator expectations. For $0 \leq \alpha \leq \alpha_0$,

$$A_\alpha = (\alpha/\alpha_0) A_{\alpha_0} + (1 - \alpha/\alpha_0) I.$$

Expanding its third tensor power gives a nonnegative combination of $A_{\alpha_0}^{\otimes k} \otimes I$ on subsets of the copies. The expectation of each term on a Schmidt-rank-two vector is a sum of nonnegative expectations, obtained by resolving the spectator systems in local product bases; each resulting vector still has Schmidt rank at most two. This proves the entire interval in Theorem 1.

## 5 Finite exact checks and the accompanying data

The computational proof uses two data files, treated as untrusted input:

- `certificate/certificate.json.gz`: the eight rational coefficient tensors, ideal coefficients, $\mu$, $\alpha_0$, and $\varepsilon$;
- `certificate/representations.json.gz`: the polynomial matrices, Grams and multiplicities defining (11)–(12).

Permutation tuples are lexicographically ordered. Subset masks use bit $i - 1$ for physical site $i$. The checker reconstructs (5) and the symmetrized auxiliary tensor from (4), checks (8) coefficient by coefficient, validates every ideal term and coefficient symmetry, and verifies all polynomial identities and positivity conditions above. It checks 16,128 polynomial diagram products in the one-site blocks and all 238 strict-margin conditions (15).

All final positivity tests are on symmetric integer matrices obtained by clearing positive rational denominators. They pass two exact tests: fraction-free symmetric elimination and signs of characteristic-polynomial coefficients. The latter is valid because, for a real symmetric $R$, nonnegative coefficients of $\det(tI + R)$ exclude a positive root in $t$, and hence exclude any negative eigenvalue of $R$. Positive Gram limits are verified separately. No floating-point eigenvalue or optimizer status is used by the checker.

From the extracted package directory, a complete replay is:

```
python -m pip install -r certificate/requirements.txt
python -B certificate/verify.py
```

The checker resolves its two data files relative to itself. The recorded replay passes the coefficient and polynomial checks and all 238 blocks by both exact methods. Its analytic input is precisely Lemma 2, proved below; the determinant and dimension arguments have been proved above. These together establish Theorem 1.

The SHA-256 digests, fixing the exact electronic data, are

`certificate.json.gz:`
`f00440126b5797ca03acc784411873cd15a39652b7dc9674dbb0a1afd759213c`

`representations.json.gz:`
`e7652dfef493f38d6ae176e62281b869e310e667480e9c0520cc1ef617f8232f`

The rational value $\alpha_0$ is a certified parameter, not an optimality claim. The strict margin in (15) is a block-coordinate margin, not a dimension-independent lower bound on $Q_{\alpha_0}(X)/\|X\|_F^2$.

# A Proof of the analytic rank-two inequality

**Lemma 2** (A rank-two three-factor inequality)**.** *Let $X$ act on a finite-dimensional complex tensor product $\mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_3$ and have rank at most two. All partial traces are unnormalized. Put*

$$N = \|X\|_F^2, \quad T = |\operatorname{Tr} X|^2, \quad B_j = \sum_{|E|=j} \|\operatorname{Tr}_E X\|_F^2, \quad g = (\sigma_1 - \sigma_2)^2,$$

*where a missing singular value is zero. Then*

$$\frac{17}{8}N - B_1 + \frac{1}{2}B_2 - \frac{5}{16}T \geq \frac{g}{4}.$$

*Proof.* Write

$$q(Y) = \sum_{E \subseteq \{1,2,3\}} (-1/2)^{|E|} \|\operatorname{Tr}_E Y\|_F^2.$$

For every square matrix, $\operatorname{Re} Y = (Y + Y^\dagger)/2$ and $\operatorname{Im} Y = (Y - Y^\dagger)/(2i)$. We first prove a positive-matrix bound, then use four positive secants and a real-Hilbert-space estimate. All identities below remain valid when a singular value vanishes; the zero matrix is immediate.

*Positive rank-two matrices.* For $P = W\tau W^\dagger \succeq 0$ of trace one, take $W^\dagger W = I_2$ and $\tau = \operatorname{diag}(\lambda, \mu)$, $\lambda + \mu = 1$. Let $\varsigma_1, \varsigma_2, \varsigma_3$ be the Pauli matrices, $K_\nu = \sqrt{\tau}\varsigma_\nu\sqrt{\tau}$, and $M_{i\nu} = \operatorname{Tr}_{\widehat{i}}(WK_\nu W^\dagger)$, where $\widehat{i} = \{1,2,3\} \setminus \{i\}$. In the unit vector $\sum_a \sqrt{\tau_{aa}}|a\rangle W|a\rangle$, the expectation of $\varsigma_\nu^T \otimes \operatorname{sign}(M_{i\nu})$ is $\|M_{i\nu}\|_1$. Partition the nine pairs $(i,\nu)$ into the three triples with $\nu - i$ fixed modulo three. In each triple these Hermitian contractions anticommute. Thus their squared expectations sum to at most one: indeed $(\sum_j t_j L_j)^2 \preceq \sum_j t_j^2 I$ for anticommuting contractions. Consequently $\sum_{i,\nu} \|M_{i\nu}\|_1^2 \leq 3$. For Hermitian $M$, its positive and negative parts give $2\|M\|_F^2 \leq \|M\|_1^2 + (\operatorname{Tr} M)^2$. Since $\operatorname{Tr} K_\nu$ is zero except for the value $\lambda - \mu$,

$$\sum_{i,\nu} \|M_{i\nu}\|_F^2 \leq \tfrac{3}{2}[1 + (\lambda - \mu)^2] = 3\operatorname{Tr}\tau^2.$$

Pauli completeness $\sum_{\nu=0}^3 (\varsigma_\nu)_{ab}(\varsigma_\nu)_{cd} = 2\delta_{ad}\delta_{bc}$, with $\varsigma_0 = I$, gives

$$\sum_{\nu=1}^3 \|M_{i\nu}\|_F^2 = 2\|\operatorname{Tr}_i P\|_F^2 - \|\operatorname{Tr}_{\widehat{i}} P\|_F^2.$$

Substitution in $q(P)$ and homogeneity therefore prove

$$q(P) \geq \frac{2\|P\|_F^2 - (\operatorname{Tr} P)^2}{8} = \frac{\operatorname{gap}(P)^2}{8}. \tag{A1}$$

Here $\operatorname{gap}(P)$ is the difference of its two eigenvalues, padding by zero.

*Four positive secants.* Take $X = USV^\dagger$, $U^\dagger U = V^\dagger V = I_2$, with $S$ diagonal and nonnegative. Set

$$s = \operatorname{Tr} S, \quad P = USU^\dagger, \quad Q = VSV^\dagger, \quad f = \operatorname{Tr}(PQ), \quad G = 2f - T; \qquad g = 2N - s^2.$$

Let $F_i$ swap receiver $i$ between two physical replicas, $\Pi = \prod_i \frac{I - F_i}{2}$, and let $F_\ell$ swap two logical qubits. Define

$$\Delta = \sum_{i,\nu=1}^3 \left\| \operatorname{Tr}_{\widehat{i}}\left[U\sqrt{S}\varsigma_\nu\sqrt{S}U^\dagger - V\sqrt{S}\varsigma_\nu\sqrt{S}V^\dagger\right] \right\|_F^2,$$

$$L = \Pi(U\sqrt{S} \otimes V\sqrt{S}), \qquad \mathcal{S} = \mathrm{Tr}\left[\frac{I + F_\ell}{2} L^\dagger L\right].$$

Both quantities are nonnegative. For $z \in \{1, -1, i, -i\}$ put $P_z = (U + zV)S(U + zV)^\dagger$ and $e(P_z) = q(P_z) - \mathrm{gap}(P_z)^2/8 \geq 0$ by (A1). The following exact identity is useful:

$$24q(X) - 4g + G = \sum_{z \in \{1,-1,i,-i\}} e(P_z) + \Delta + 64\mathcal{S}. \tag{A2}$$

For completeness, it can be checked with three short expansions. Put $E = q(P) + q(Q)$ and define

$$h = \sum_J (-1/2)^{|J|} \mathrm{Tr}[(\mathrm{Tr}_J P)(\mathrm{Tr}_J Q)], \qquad c_j = \sum_{|J|=j} \mathrm{Tr}(P_J Q_J),$$

where $P_J = \mathrm{Tr}_{J^c} P$ and similarly for $Q_J$. Pauli completeness and expansion of the two projectors give, respectively,

$$\begin{aligned} 8h &= 8f - 4c_2 + 2c_1 - s^2, \\ 4E + \Delta &= 8N - s^2 - 4B_1 + 2c_1, \\ 64\mathcal{S} &= 4(s^2 + N - c_1 + c_2 - f - B_1 + B_2 - T). \end{aligned}$$

For the middle equality the cross Pauli sum at receiver $i$ is $2\| \mathrm{Tr}_i X\|_F^2 - \mathrm{Tr}(P_i Q_i)$. For the last equality the identity and logical-swap contributions are $(s^2 - c_1 + c_2 - f)/8$ and $(N - B_1 + B_2 - T)/8$. Finally, four-phase averaging gives $\sum_z q(P_z) = 4E + 8h + 8q(X)$ and $\sum_z \mathrm{gap}(P_z)^2 = 16g + 8G$. The latter follows by applying $\mathrm{gap}(M)^2 = 2\,\mathrm{Tr}\, M^2 - (\mathrm{Tr}\, M)^2$ to the two-by-two Gram matrices $2S + zZ + \bar{z}Z^\dagger$, $Z = \sqrt{S}U^\dagger V\sqrt{S}$. These expansions prove (A2), hence

$$q(X) \geq (4g - G)/24. \tag{A3}$$

*Hermitian reduction and real Bessel inequality.* For positive $A, B$ with $AB = 0$, positivity of $\prod_i (I - F_i)$ implies $b \leq u + a$, where $u = \mathrm{Tr}\, A\, \mathrm{Tr}\, B$, $a = \sum_i \mathrm{Tr}[(\mathrm{Tr}_i A)(\mathrm{Tr}_i B)]$, and $b$ is the analogous sum over two traced factors. Thus their polarized $q$ value is $-a/2 + b/4 - u/8 \leq u/8$. Apply this and (A1) to the positive and negative parts of a Hermitian $Y$ with at most two eigenvalues of either sign, obtaining

$$q(Y) \geq [2\|Y\|_F^2 - \|Y\|_1^2]/8.$$

Both $\mathrm{Re}\, X$ and $\mathrm{Im}\, X$ have this inertia bound: their quadratic forms vanish on $\ker X$, a subspace of codimension at most two. Since $q(X) = q(\mathrm{Re}\, X) + q(\mathrm{Im}\, X)$,

$$q(X) \geq \frac{2N - \|\mathrm{Re}\, X\|_1^2 - \|\mathrm{Im}\, X\|_1^2}{8}. \tag{A4}$$

Put $C = U^\dagger V$. Choose a Hermitian unitary $J = \mathrm{sign}(\mathrm{Re}\, X)$, assigning signs on its kernel, and a Hermitian Frobenius-orthonormal basis $E_0, \ldots, E_3$ of $M_2$. For the real inner product $\mathrm{Re}\,\mathrm{Tr}(A^\dagger B)$, the five matrices $JU/\sqrt{2}, iUE_0, \ldots, iUE_3$ are orthonormal. Indeed their cross inner products vanish because $U^\dagger JU$ and the $E_\mu$ are Hermitian. Bessel's inequality on $VS$ gives

$$\tfrac{1}{2}\|\mathrm{Re}\, X\|_1^2 + \|\mathrm{Im}(CS)\|_F^2 \leq N.$$

Exchange $U, V$ and average. The elementary entrywise identity

$$\|\mathrm{Im}(CS)\|_F^2 + \|\mathrm{Im}(SC)\|_F^2 - 2\|\mathrm{Im}(\sqrt{S}C\sqrt{S})\|_F^2 = \tfrac{1}{2}\sum_{i,j}(S_{ii} - S_{jj})^2 |C_{ij}|^2 \geq 0$$

yields $\|\mathrm{Re}\, X\|_1^2 \leq 2N - 2\|\mathrm{Im}\, Z\|_F^2$. Apply the same argument to $iX$ and add, using $f = \|Z\|_F^2$:

$$\|\mathrm{Re}\, X\|_1^2 + \|\mathrm{Im}\, X\|_1^2 \leq 4N - 2f.$$

Equation (A4) now gives $q(X) \geq (G + T - 2N)/8$. Take one quarter of this bound and three quarters of (A3):

$$q(X) \geq -\frac{N}{16} + \frac{T}{32} + \frac{g}{8}.$$

Since $2q(X) + N/8 - T/16 = 17N/8 - B_1 + B_2/2 - 5T/16$, the claim follows. If the entire physical space is one-dimensional, it is immediate directly; otherwise rank-one SVDs can be padded to the two-column isometries used above. □